
IIO-13150 Modeling and Optimization I

Credits

6.0 (3 hours lecture)

Lead Instructor

Miguel de Lascurain Morhan

Textbook

Investigación de Operaciones (10a ed.). Hillier, Frederick y Lieberman, Gerald. McGraw-Hill, 2015.

Supplemental materials

Operations Research - An Introduction". (9th ed.). Taha, Hamdy A. Prentice Hall, 2011.

Operations Research Applications and Algorithms, (3rd ed.) Winston W. Prentice Hall Hispanoamericana Duxbury Press, 2004.

Course Description

Principles and training required to develop mathematical models to support decision making in different Engineering linear problems.

The theory and solution methods related to linear programming problems, including integer, binary, mixed and dynamic programming applications.

The use of computer packages in order to solve linear programming problems.

Prerequisites

Differential and Integral Calculus II (MAT 14101).

Linear Algebra I (MAT 14201)

Major Status

Required

Course Outcomes

1. To apply basic linear algebra to solve linear programming problems.

2. To generate different solutions to a problem and to compare them to the optimal one.
3. To develop mathematical models for a large variety of systems.
4. In class discussion helps to analyze and develop decision models.
5. To learn how to recognize and solve problems with constraints.
6. To learn how to use computer packages to solve linear programming problems.

ABET Outcomes

- a) An ability to apply knowledge of mathematics, science and engineering.

Course Topics

- Linear programming problems and the Simplex Method.
- Primal and dual problems.
- Sensitivity analysis.
- Integer programming and the Branch and Bound Method.
- Solving the transportation problem using MODI.
- The assignment problem.
- Binary programming and the Balas Method.
- Network problems.
- Dynamic programming.